



Outcome Evaluation Protocol

Version 7.0

Version History (V7 Changes Highlighted in Yellow Throughout)

V7.0 (February 2026): Clean restart with flipped cohort. Key changes from V6:

- Research question updated to include providing WQ readings as part of intervention
- Hypothesis simplified to remove "and these are implemented" (per Jennifer's comment)
- Farm assignment: 25 Treatment, 28 Control (44 of these from previous cohort)
- Follow-up timing: TRIAD system with Day 2 + Day 3 follow-ups collected
- Unit of analysis clarified: pond-per-day (not per-measurement)
- Self-initiated actions recorded once per event only (to avoid duplicates)
- Updated timelines: February - May 2026 (16 weeks)
- Added Annexure A: Suggested Protocol Additions (control farmer survey, monsoon reachability, summer considerations)

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1. Introduction

The current core program of Fish Welfare Initiative (FWI) is the Alliance for Responsible Aquaculture (ARA). By joining the ARA, farmers commit to reduced stocking densities and to maintaining water quality in required ranges. ARA field teams collect water quality data—including dissolved oxygen (DO), pH, and ammonia—from member farms approximately once a month and provide the farmers with recommendations for corrective actions in the event of measurements revealing that key water quality parameters are out-of-range, indicating that fish may be exposed to unhealthy conditions.

While the ARA program includes components on stocking density and general farmer engagement and support, the core intervention is the provision of water quality readings and corrective actions that aim to improve water quality. The underlying assumption behind this intervention is that if water quality is poor, fishes suffer, so by alerting farmers to water quality issues and providing them with recommendations to improve water quality, the welfare of fishes improves.

While the individual corrective actions issued by ARA field teams may be based on available evidence, there is currently a lack of evidence to show that the intervention (i.e. provision of water quality readings and corrective actions that aim to improve water quality) actually improves water quality. Notably, there is only limited information about the counterfactual: what happens to water quality in the absence of corrective actions? ARA field teams conduct follow-up visits after corrective actions have been recommended to check if water quality has returned to normal, but that is not sufficiently rigorous evidence to show the correction action(s) actually caused the improvement. Water quality is inherently dynamic, and can change over time.

To provide rigorous evidence to show that the ARA, through the provision of water quality readings and corrective actions, actually improves water quality, a controlled study will be conducted, comparing changes in water quality at farms at which corrective actions are implemented to changes in water quality at farms at which no corrective actions are implemented. The intention is to conduct an effectiveness study of corrective actions, not an efficacy study (Singal, 2014)¹. The **effectiveness study** will evaluate whether the corrective actions recommended to farmers under normal circumstances—as opposed to rigid experimental conditions—leads to improved water quality. With respect to our evidence-based program funnel, this study is a field-test.

This effectiveness study serves two key purposes. Firstly, it serves to evaluate if the intervention actually improves water quality. By evaluating the effectiveness of corrective

¹ Singal, A.G., Higgins P.D.R. and Waljee A.K. Primer on Effectiveness and Efficacy Trials. Clinical and Translational Gastroenterology (2014) 5, e45; doi:10.1038/ctg.2013.13



actions, this study also serves as an **outcome assessment** of the ARA. The ARA's intended impact is to improve the welfare of fishes. Assessing changes in welfare is a considerable challenge, which complicates conducting a true impact assessment of the ARA. However, to achieve this desired impact, the ARA's intended outcome is to improve water quality; if this outcome (improved water quality at fish farms) is not successfully achieved, then the intended impact (improved fish welfare) cannot be achieved. As such, this outcome assessment is considered a critical evaluation to provide evidence that the ARA is making a difference to the lives of fishes.

1.1 Hypothesis

Ponds that experience out-of-range water quality instances, **are provided with water quality readings and** corrective actions will have more frequent water quality improvements (back into required ranges) than ponds who do not receive water quality readings, nor corrective actions.

Note: V7 removes "and these are implemented" from the hypothesis. We are testing the full ARA process, including whether farmers choose to implement CAs. Cases where farmers do not implement CAs are informative about the ARA process and are included in the intention-to-treat analysis.

1.2 Research Question

Does providing farmers with water quality readings and corrective actions result in improved water quality?

2. The Intervention to be Evaluated

Following is the process of improving water quality through the provision of water quality information and corrective actions. Figure 1 further illustrates the individual steps.

- **Regular Measurement:** ARA staff visit farmers at least every four weeks for a water quality measurement and to assess further welfare indicators. Staff will follow the ARA SOPs for these measurements. **SOP: statement of purpose**
- **Poor WQ Detection:** If one or more of the parameters fall(s) outside of the ARA's required ranges, staff prescribe corrective actions to the farmer.
- **Corrective Action Implementation:** The farmer is expected to implement corrective actions as soon as possible and according to ARA staff's instructions.
- **Follow-Up Measurement(s):** ARA staff re-visit the farm on the second day of the corrective action implementation to check whether a) the farmer implemented the corrective actions and b) the targeted water quality parameter improved. These

Needs
rewording



follow-up visits recur until the targeted water quality parameter improved into the required ranges again.

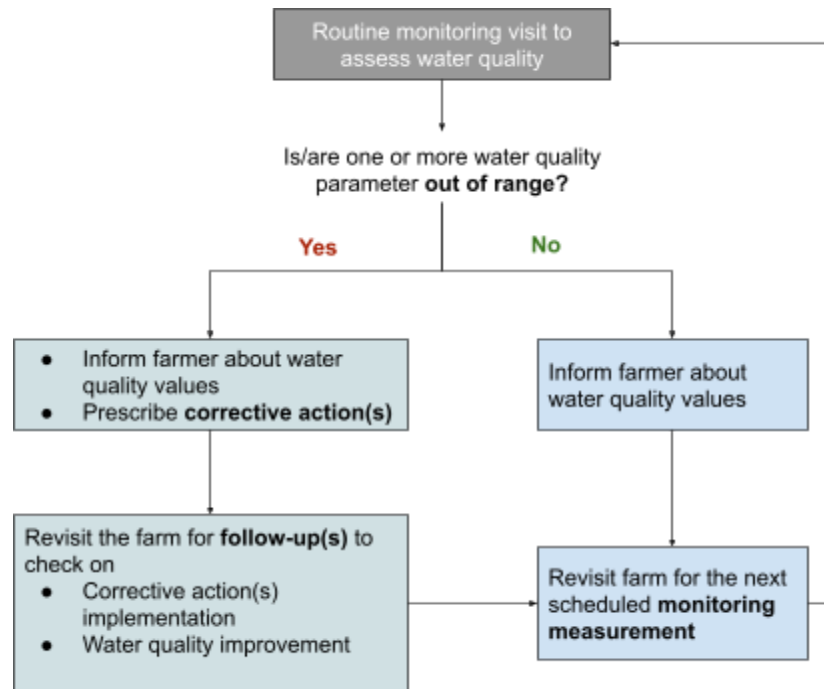


Figure 1. Overview of how the ARA helps farmers to improve water quality

2.1 Water Quality Parameters

During regular and follow-up measurements we collect three main water quality parameters which we identified to affect fish welfare and be outside appropriate ranges most frequently.

For any measurement, we visit the farm twice within one day. One time in the morning and one time in the evening. We adjust the visit times seasonally to account for the change in sunrise and sunset times throughout the year. The morning visit is between 6-8am in summer and 6.30-8.30am in winter. The evening visit is between 4-6pm in summer and 3.30-5.30pm in winter.

During the morning measurement, staff collect two water samples from a farm: one DO sample and one ammonia sample. They also measure pH and temperature from a sample bottle in the field. The DO sample is fixed with Win A and Win B after collection. Both the DO and ammonia samples are then taken to the lab for analysis.

Water Quality Parameter	Required Range	Method	Collection
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Dissolved Oxygen (Morning)	3 - 5 mg/L	Winkler's Method (from sample)	Morning
Dissolved Oxygen (Evening)	8 - 12 mg/L		Evening
pH	6.5-8.5	pH pen	Morning & Evening
Temperature	/ (used for ammonia)	Thermometer	Morning & Evening
Ammonia	<0.05 mg/L	API Test Kit (from sample)	Morning only

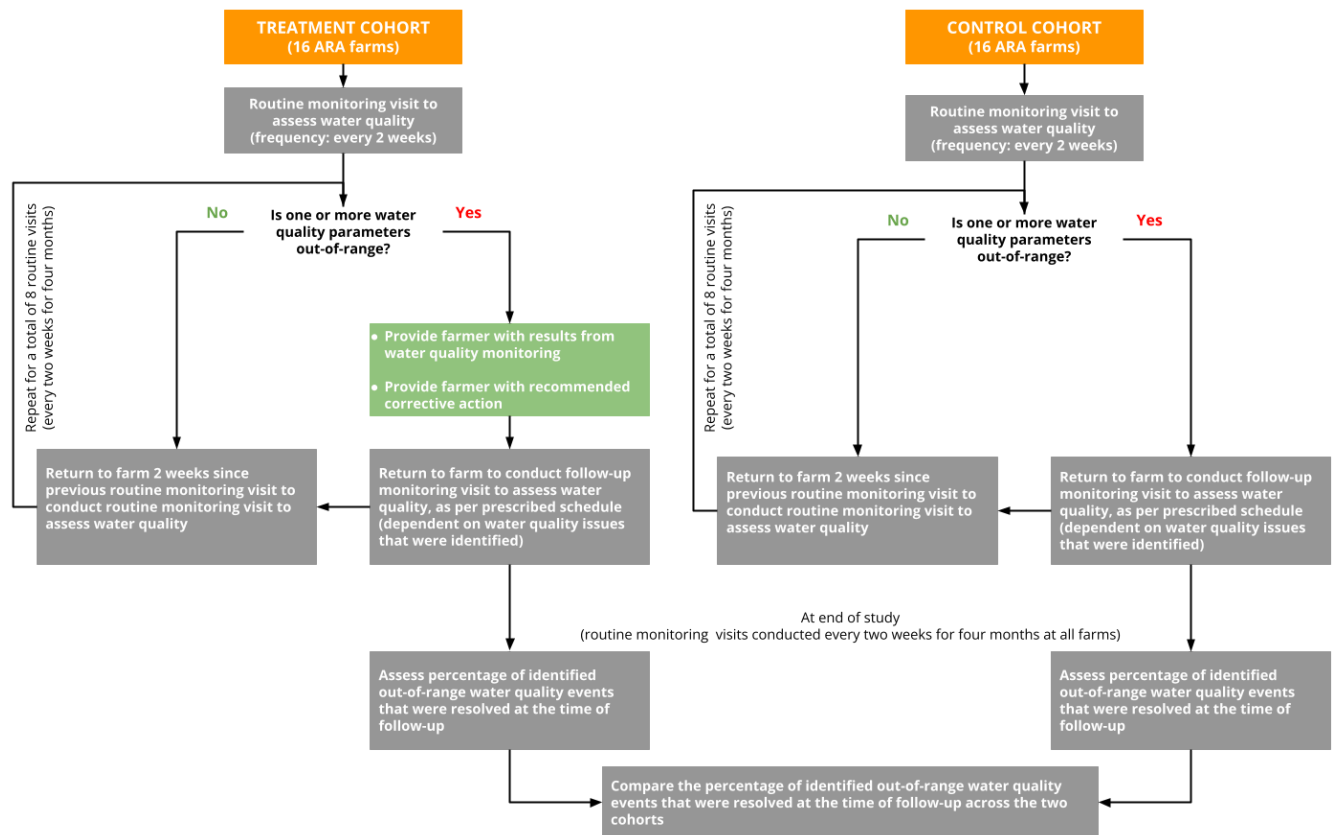
Table 1. The required water quality ranges farmers commit to when joining the ARA.

The accuracy of these measurements is verified through [comprehensive assessments](#) and [continuous monitoring protocols](#).

2.2 Corrective Actions

Upon sample analysis at the lab, ARA staff call the farmer to inform them about corrective action(s) or wait until the evening visit to meet them in person—depending on how urgent the corrective action is.

The relevant corrective actions are outlined in [these SOPs](#) and differ for each water quality parameter and whether it is above or below the range.



3. Experimental Design

The principal indicator for assessing the effectiveness of the ARA's corrective actions is improved water quality: if corrective actions improve water quality, we are looking to find a statistical difference in the number of follow-up visits that result in improved water quality at farms that have regular water quality monitoring combined with the issuance of corrective actions (aka treatment group) compared to farms that only receive regular water quality monitoring without being told these, nor corrective actions (aka control group).

The treatment and control groups will both comprise farms currently part of the ARA. With this, we accept that there has likely been a change in behaviour in both groups because they will already know about our corrective actions and specific dosages we advise. We believe that taking ARA farms as controls is better than choosing non-ARA controls because

- **Logistics:** There are insufficient non-ARA farms left in our programming area to be recruited as controls. If we expand too far beyond current programming areas, farm characteristics such as water source, stocking density, or feed practices, may differ to the point of not being comparable to treatment farms.



- **Feasibility:** Recruiting non-ARA farms as controls; taking measurements but not telling them the readings; will be extremely challenging, if not impossible. These farmers have incentive to trust or work with us. We could introduce incentives (e.g., cash rewards) but that brings its own challenges by introducing another variable that may falsify the study results.
- **Honesty:** We have strong relationships with ARA farmers and they feel comfortable telling us about their farm and practices. We require control farmers to tell us about actions they took themselves to determine whether water quality improved naturally or because of something they did. ARA farmers are more likely to tell us these details, allowing us to further interpret the study results.
- **Accuracy:** ARA farmers trust us and our CAs and we can therefore be more certain in them implementing them according to the SOPs. This will allow us to receive more accurate results from this study.

Farms from these two cohorts will be matched such that they are considered similar in terms of size, location, and aquaculture practices. Figure 2 provides an overview of the study design.

As this is an effectiveness study designed to test the intervention in the context in which it would be implemented outside the confines of a study, the corrective actions will be administered by farmers—not researchers/ARA field teams—with no additional support beyond what would normally be provided as part of the ARA.

The study will run for a period of four months, after which the percentage of follow-up visits with resolved water quality issues (i.e. within acceptable thresholds for DO, pH, etc) will be compared across the two cohorts. If the corrective actions improve water quality, we would expect to see a higher percentage of resolved water quality issues in the treatment group.

Figure 2. Overview of study design. The key difference between the two cohorts will be that control farms will not be informed of the findings from routine monitoring visits, and no corrective actions will be recommended to farmers. In the event that a routine monitoring visit at a control farm identifies one or more water quality parameters being out-of-range, field teams will revisit that farm to assess if water quality has resolved on its own, in the absence of corrective actions.

3.1 Target Population:

The ARA currently supports farmers in two districts of Andhra Pradesh: Nellore and Eluru. This study will focus only on farms in Eluru because this is where the program is expanding



and because Nellore is less densely populated with farms and thus doesn't offer sufficient farms for running this study. The villages in which the ARA operates are marked in purple [here](#).

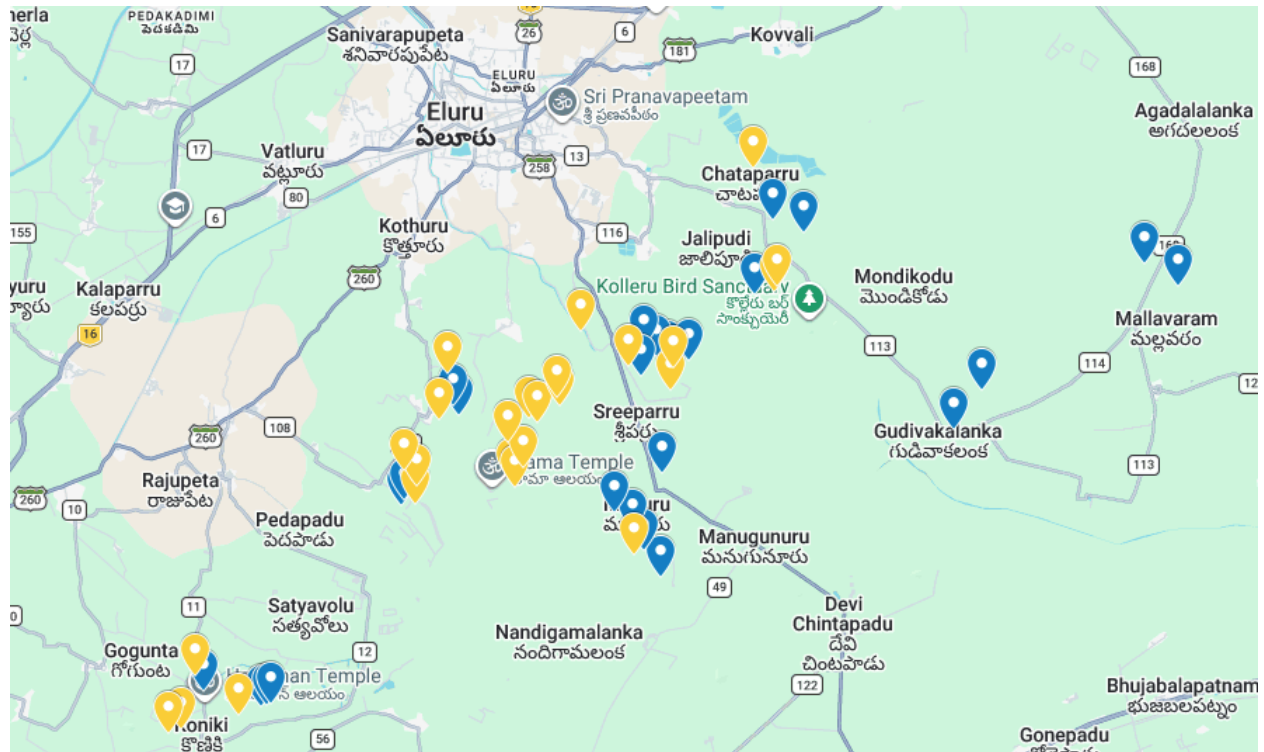


Figure 3. Overview of the participating treatment (blue) and control farms (yellow).

Farms in the Eluru district have bigger farm sizes (~14 acres on average), don't tend to do pond preparation, and most of them only have access to seasonal canal water which tends to be polluted with nutrients. As a result, Eluru farms tend to experience more water quality issues and have difficulty addressing these given the vast size of many farms there.

4. Methodology

4.1 Sample Selection

4.1.1 Number of Farms to Include in Study

Based on the OpenEpi tool using the Fleiss methodology (w/ continuity correction), we require 32 sample points (i.e., out-of-range events) to gain statistical significance—16 OOR events at treatment farms and 16 at control farms.



Sample Size: X-Sectional, Cohort, & Randomized Clinical Trials			
Two-sided significance level(1-alpha):			95
Power(1-beta, % chance of detecting):			80
Ratio of sample size, Unexposed/Exposed:			1
Percent of Unexposed with Outcome:			37
Percent of Exposed with Outcome:			89
Odds Ratio:			14
Risk/Prevalence Ratio:			2.4
Risk/Prevalence difference:			52
	Kelsey	Fleiss	Fleiss with CC
Sample Size - Exposed	14	13	16
Sample Size-Nonexposed	14	13	16
Total sample size:	28	26	32

Figure 4. The required sample size for this study to get statistically significant results is 32 out of range measurements.

V7 CLARIFICATION: Unit of Analysis

The unit of analysis is POND-PER-DAY, not per-measurement.

This means:

- If a pond has OOR readings in both morning AND evening on the same day, this counts as ONE OOR event (not two)
- Resolution is assessed at the pond-day level: did the pond return to acceptable ranges?
- Each valid event includes an initial OOR event and a day 3 follow-up.
 - Day 2 follow-ups are supplemental and used to further analyze how quickly CAs show results.

V7 Farm Assignment

This study will include 53 farms in total:

- 25 Treatment farms
- 28 Control farms

This assignment is based on flipping the control/treatment status from the previous study phase (Sep-Dec 2025), with farms that have become unavailable (due to harvest



or other reasons) excluded. An additional 6 ponds from the data campaign have been added to the control farms (which was at 19 total after flipping).

Change from V6: V6 had 54 farms (27 treatment + 27 control). V7 has 44 farms (25 treatment + 19 control).

Our unit of analysis for this study are OOR events (pond-per-day) where at least one WQ parameter is OOR. Due to variable stocking and harvest times, farms may be phased in and out of the study throughout the study period.

4.1.2 Selection Criteria

Farms enrolled in the study must meet the following criteria:

- **Location:** Only farms in the Eluru district of Andhra Pradesh will be considered to ensure homogeneity of environmental conditions, logistical feasibility, and comparability of aquaculture practices.
- **Life cycle:** Only grow-out farms for Indian major carp (IMC) will be eligible. All farms should be within the first 4 months of the grow-out process at the time of enrollment. Farms nearing harvest or practicing partial harvest will be excluded.

4.1.3 Matching Criteria

[Selected farms](#) were randomly assigned to control and treatment groups. After assignment, we evaluated the groups on these criteria:

- Pond size
- Stocking densities
- Village
- Fish weight
- % WQ OOR

The random assignment seemed to have distributed farms mostly evenly and we thus didn't change any assignment after our review.

4.1.4 Process to Select Farms

For V7, the cohort is derived from flipping the treatment/control assignment from the [previous study phase](#). Farms that have harvested or become otherwise unavailable have been excluded.

4.2 Data Collection



All data will be collected by ARA Eluru staff who are fully trained on the relevant protocols. The ARA focus pond grouping will be temporarily paused throughout the study to allow sufficient time for data collection at participating farms.

Unless otherwise stated, regular [ARA protocols](#) will be followed for data collection and entry throughout the study.

4.2.1 Regular Monitoring Visits

Each farm will be visited every two weeks (twice a month for four months, so eight times over the course of the study). Two weeks is defined as one visit every fourteen days and the frequency will be based on regular visits (**not** the follow-up visits).

At every farm, the parameters in table 2 will be assessed:

General Info	Water Quality	Behavioural Welfare Indicators	CA Information
Date & Time	Dissolved Oxygen	Air gulping	CA requested <i>(treatment only)</i>
Pond ID	pH	Tail splashing	Amount requested <i>(treatment only)</i>
Equipment	Ammonia	Mortalities	Self-initiated actions taken
Weather			
Staff Name			
Follow-Up (Yes/No)			
Feed amount			
Fish weight			
Stocking density			

Table 2. Parameters to be measured during routine monitoring visits.

Treatment farms will receive information about their WQ values and CAs (if necessary).



Control farms will NOT receive any information and thus will not get the WQ readings nor CAs.

4.2.2 Corrective Actions

In the event that any WQP is measured to be out of range, the following process will be followed:

- **Treatment farms:** the [appropriate corrective action](#) will be given to the farmer per the [SOPs](#).
- **Control farms:** No corrective action will be communicated to the farmer, and no water quality readings will be shared with the farmer.

4.2.3 Follow-Up Monitoring Visits After Issuance of Recommended Corrective Action

For every out-of-range event detected, follow-up visits to the farm will be conducted to assess if the specific water quality issue has been resolved.

V7 FOLLOW-UP PROTOCOL

A valid data point requires a follow-up on day 3.

We are also collecting day 2 follow-ups for comparison of day 2 and day 3 differences.

- **Initial OOR Measurement (Day 0)**
- **Follow-Up on Day 2** (secondary comparison point)
- **Follow-Up on Day 3** (primary comparison point)

Day 3 is the PRIMARY outcome measure for treatment vs. control comparison, aligned with current ARA operational practice.

Day 2 is collected additionally to enable comparison of follow-up timing effects and safeguard against inconclusive results.

TRIAD Illustration: Weekly Calendar Example

Example: OOR detected on Monday



Monday	Tuesday	Wednesday	Thursday
DAY 0	DAY 1	DAY 2	DAY 3
Initial OOR detected (TRIAD Part 1)	Farmer implements CA (Treatment only)	Follow-Up #1 (TRIAD Part 2) Secondary	Follow-Up #2 (TRIAD Part 3) PRIMARY

Example: OOR detected on Thursday

Thursday	Friday	Saturday	Sunday
DAY 0	DAY 1	DAY 2	DAY 3
Initial OOR	CA implemented	FU Day 2 (Weekend - still required!)	FU Day 3 (Weekend - still required!)

IMPORTANT: Follow-ups will be conducted on Day 2 AND Day 3 regardless of whether that day is a weekend or holiday. On these days, one staff will cover all relevant farms and receive a compensatory leave day the following week.

Note: During regular ARA programming, we would take follow-ups until water quality improved, but for the purposes of this study, we are assessing only the Day 2 and Day 3 follow-ups. This avoids artificially inflating success rates and captures the speed or likelihood of improvement under normal, short-term corrective measures.

During follow-up visits, ARA staff will collect the parameters in Table 3.

General Info	Water Quality	Behavioural Welfare Indicators	CA Information
Date & Time	DO	Air gulping	Actions implemented



			(yes/no and which ones)
Pond ID	pH	Tail splashing	Implementation date
Equipment	Ammonia	Mortalities	Details/amount
Weather			If the farmer didn't apply corrective actions, what were the reasons?
Staff Name			Self-initiated actions taken
Follow-Up (Y/N)			Targeted water quality parameters improved (Y/N)
Feed amount			
Fish weight			
Stocking density			

Table 3. Parameters to be measured during follow-up visits.

**Note (per Jennifer): "Targeted WQ parameters improved" will be entered by associates for control farms in the data tool.*

V7 CHANGE: Self-Initiated Actions Recording Adjustment

Self-initiated corrective actions should be recorded only once per event (either in the morning OR evening entry, not both). This prevents duplicate values in the dataset. Record in the morning entry unless information is only obtained during the evening visit.

4.2.4 Weather Data

We expect weather—and affiliated factors like water temperature—to be one of the primary drivers for water quality changes. We will thus monitor weather closely to include eventual changes in the final data analysis.



- Staff will note down observed weather and rainfall during their visit.
- We will gather hourly weather information from [Open Meteo](#).

Weather data will be integrated into the final analysis to explore potential confounding or mediating effects.

4.3 Data Quality Control and Assurance

Compliance with the study and CA SOPs is vital for the success of this study. We will closely monitor

- staff's compliance with [SOPs](#) when suggesting CAs (4.3.1)
- farmers' compliance with suggested CAs when implementing them (4.3.2)
- staff's compliance with control farm protocols (i.e., maintaining the treatment blindness of control farms) (4.3.3)

All quality control actions will be documented on the [study tracker](#).

4.3.1 Staff Compliance with Corrective Action SOPs

We have been conducting [surveys](#) to ensure that our staff are complying with the CA SOPs and will continue doing this survey once throughout the study period.

We are also analyzing the data with respect to SOP adherence. Compliance checks will continue during the study period, and deviations will be addressed through coaching or retraining if required.

4.3.2 Farmer Compliance with Corrective Action SOPs

Two mechanisms will be used to track whether farmers implemented the recommended CAs correctly:

1. Analysis Of Self-Reported Information

- During follow-up visits, we will ask farmers what specific CAs they have implemented. This information is entered on our data tool.
- The Director of Programs will review and analyze these entries to ensure that farmers implement CAs according to the SOPs. If farmers do not comply, we will either remove or account for the measurement in the analysis.

2. Shadowing & Visual Verification



- Shadowing visits will be conducted for a minimum of **25% of expected treatment OOR events, so 4 in total** (change from V6: was 25% of treatment farms).

4.3.3 Staff Compliance with Study Protocol (Control Farms)

The diligent implementation of the study protocol is crucial for accurate data. In particular, we need to be certain that the control farms are treated appropriately and Program Associates do not tell them the water quality readings, nor give corrective actions.

The risk of control farms being informed about readings is somewhat minimized by not taking all measurements in-field but further monitoring protocols are required to ensure compliance. We will thus conduct shadowing days where an ARA manager shadows ARA staff for a control farm visit and checks for:

- Staff not telling farmers WQ readings
- Staff not giving farmers CAs

We will have a minimum of two of these visits per staff conducted for regular measurements and two for follow-up measurements—all of these at control farms.

If disparities to the protocol are detected, action will be taken immediately, including possible data exclusion and refresher training for involved staff.

4.3.4 Team Meetings and Reporting

Regular reporting will ensure appropriate implementation of protocols and offer a review point for collected data. There will be two types of touchpoints:

- **Daily debriefs** — conducted only if a contamination incident occurs or if there are challenges in collecting treatment data on a given day. Led by Chaitanya. Not required on routine data collection days.
- **Monthly study progress report** — prepared by the study lead covering all periods within the month.

The 2-weekly progress meetings originally specified in this protocol were discontinued by mutual agreement between the study lead and implementation lead, as the monthly report and tracker were deemed sufficient for ongoing monitoring. This decision was made during Phase 2 implementation.

4.4 Analysis of Data

Primary Hypothesis



If corrective actions improve water quality, we would expect to see a higher percentage of resolved water quality issues in the treatment group compared to the control group.

The data will be analyzed blind by someone who was not involved in the farm selection & randomization and the study execution. Jennifer and Marco (if he's not the primary analyzer) will quality check the analysis.

A discussion of statistical significance (p-values) and whether the results support or reject the hypothesis will be performed.

- **Null Hypothesis (H_0):** There is no significant difference in the percentage of resolved water quality issues between the treatment and control groups.
- **Alternative Hypothesis (H_1):** The treatment group has a significantly higher percentage of resolved water quality issues than the control group.

Descriptive Statistics

Descriptive statistics will be performed to summarize the baseline and follow-up measures:

- Mean and standard deviation of water quality parameters across both treatment and control groups
- Proportion of resolved OOR events at follow-up in both groups (overall and by parameter)
- Distribution plots and variance homogeneity tests
- Outliers will be assessed using standardized residuals and influence diagnostics

Comparative Statistical Tests

To formally assess whether corrective actions lead to improved water quality:

- Paired t-test (data normally distributed) or Wilcoxon Signed-Rank Test (data not normally distributed) for matched data
- Independent t-test (data normally distributed) or Mann-Whitney U Test (data not normally distributed) for unmatched comparisons

Day 2 vs Day 3 Comparison (Secondary Analysis)

A secondary analysis will compare resolution rates between Day 2 and Day 3 follow-ups to determine if follow-up timing affects observed outcomes. This will help inform future protocol decisions regarding optimal follow-up timing and provides a safeguard against inconclusive results if Day 3 shows no effect but Day 2 does.

Multivariate Analysis



For a more nuanced analysis, we may use regression models to account for potential confounding variables (e.g., farm size, pond depth, baseline water quality). This may include:

- Multiple Linear Regression (for continuous outcomes)
- Logistic Regression (if the outcome is binary: resolved vs. unresolved)
- ANCOVA to compare post-treatment outcomes between groups while adjusting for baseline differences

4.5 Staffing

Task	Staff Name(s)	% of Staff Time spent on OE
Overall study lead (responsible for accurate implementation of protocol, troubleshooting, and reporting)	Jennifer	~20-50% (depends on period of study)
Implementation supervision (overseeing day-to-day implementation)	Chaitanya	60%
Data collection (regular monitoring and follow-ups)	Nagaraju Ravi Teja	~60-70%
Quality checks: Analysis Of Self-Reported Information	Jennifer	5%
Quality checks: Shadowing/Visual Verification of Staff and Farmer Compliance with Protocols	Multiple	10-20%
Data analysis	Marco or External Analyst	100% for ~two weeks

4.6 Timelines

Based on an expected frequency of 10% of routine monitoring visits identifying at least one water quality parameter to be out-of-range, fortnightly visits for a period of approximately 16 weeks should result in sufficient data to allow for comparison between the two cohorts.



V7 Timeline

- **Field work commencement:** Week of February 3, 2026 (exact date TBD)
- **Expected field work completion:** Approximately May 18, 2026 (~16 weeks)
- **Data analysis:** May-June 2026
- **Results publication:** End of Q2 or early Q3 2026

Target: 16 OOR events for treatment and 16 OOR events for control (32 total valid TRIADs). Study may finish earlier if targets are met, or extend beyond May 17, if needed.

5. Limitations

While we try to mitigate bias and noise as much as possible, this study is field-based and thus comes with a number of limitations. We should take these into account when interpreting the study results.

- **Participating Farms:** Using ARA farms for this study comes with benefits but also presents the challenge of studying farms that potentially may have better WQ overall or otherwise modified their pond environment in ways that improves WQ differently than for non-ARA farms. E.g., if the ARA made them more aware of the importance of WQ and they now regularly apply probiotics, their water buffering capacity may simply be higher and OOR measurements may naturally improve more quickly or CAs may naturally be more effective. Given that we use ARA farms for the control and treatment cohorts some of this noise is accounted for but we are not getting reliable information on how this study would fare with non-ARA farmers in one or both cohorts.
- **Sample Selection:** While we take necessary measures to adjust match control and treatment farms appropriately, there is only a limited number of farms in the ARA and Eluru and we therefore may have to make small compromises. We will not be able to find a 100% perfect match for selected farms.
- **Limited Geographic Representativeness of Farms:** Our Eluru ARA farms are likely *not* [representative](#) of other regions in AP or India. However, it is unclear whether any farming region is. It appears that farming clusters adopt similar practices which are not common elsewhere and finding a truly representative region is thus difficult. The variability in farm characteristics (e.g., farming practices, ecological conditions, and farmer behavior patterns) has implications for how applicable the study results are to regions outside of Eluru.
- **CA SOP Adherence:** Our ability to check whether farmers implement CA per our SOPs is limited. While we take some [initiative](#) to ensure this happens, there is a



remaining risk of farmers reporting (intentional or unintentional) the implementation of CAs wrongly.

- **Parameter-Dependent Results:** DO makes up ~60-90% of OOR measurements (depending on the season) and we thus expect most CAs to be targeting DO. The study results may thus be less descriptive of the effectiveness of pH and ammonia CAs.
- **Limited Follow-Up Duration:** This study assesses only short-term (48–72h) changes in WQP. Long-term effectiveness of CAs on pond ecology and fish welfare remains unmeasured.
- **Control Group Contamination Risk** Despite best efforts to keep control farmers blind to WQ results and CAs, there is a risk of information spillover, especially if neighboring farmers are in different groups and communicate regularly.

What Study Tells Us And What It Doesn't

What It Tells Us	What It Does Not Tell Us
Whether WQ improves between ARA farms if we give WQ readings and CAs vs. if we don't.	Whether treatment farm improvements are solely due to CAs or also other factors.
Whether WQ improves quickly and short-term (within 2-3 days after the parameter being OOR).	Whether WQ would improve/deteriorate beyond 3 days and how it develops long-term.
Whether there is a difference in how quickly CAs improve WQ (i.e., Day 2 vs Day 3 effect).	Farmer behaviour change: Does telling farmers WQ readings or the issuance of CAs (or both) ultimately nudge farmers to improve water quality?



Annexure A: Suggested Protocol Additions

This annexure contains suggested additions that the team may consider incorporating into the protocol. These are presented as options rather than requirements, allowing flexibility in implementation based on operational capacity and study priorities.

A.1 Control Farmer Survey: Independent WQ Testing

A key unknown from Phase 1 was whether control farmers were obtaining water quality information through means other than our visits. Understanding this would help interpret results and assess the true counterfactual. The following survey questions are suggested:

- "Have you tested water quality in the past month through any means other than our visits?"
- "What made you decide to implement [specific action] on [date]?" (asked when self-initiated actions are reported)
- "Do you discuss water quality management with neighboring farmers?"
- "Have you received water quality information from government, other NGOs, or other sources?"

Timing Options: The team should decide when these questions are asked. Options include: (A) at enrollment only, to establish a baseline understanding; (B) periodically during regular visits (e.g., monthly), to track changes over time; or (C) only when self-initiated actions are recorded, as a targeted follow-up. Each approach has trade-offs: baseline-only is simpler but misses changes; periodic adds burden but captures dynamics; action-triggered is efficient but may miss farmers who test WQ but take no action. A combination approach (baseline + action-triggered) may offer the best balance.

A.2 Summer-Specific Considerations (February-May)



The V7 study period (February-May 2026) coincides with summer conditions in Eluru. While this offers advantages over monsoon conditions, there are specific considerations to be aware of:

Advantages:

- Minimal rainfall improves follow-up timing compliance
- More predictable scheduling and road accessibility
- Reduced risk of weather-related data loss

Considerations and Risks:

1. Water temperature effects: Higher water temperatures reduce DO saturation capacity and affect ammonia toxicity calculations. OOR patterns may differ from monsoon-season baseline data (potentially more morning DO issues due to warmer nights increasing overnight respiration).

2. Water level effects: Pond water levels may drop significantly during summer, leading to higher fish concentration per water volume. This may trigger early harvests, increasing dropout risk. Consider monitoring water levels as an early warning indicator for potential dropouts.

3. Diurnal DO patterns: Morning DO is likely to be lower than during monsoon season due to warmer nights (higher respiration rates). The 10% OOR rate assumption (based on monsoon data) may not hold; rates could be higher or lower depending on conditions.

4. Field staff considerations: Morning visits (6-8am) should be manageable, but afternoon visits (4-6pm) may be challenging during peak summer heat. Staff hydration and safety should be monitored.

5. Seasonality effects on interpretation: Results from a summer study may not be directly comparable to monsoon-season data. CA effectiveness may vary by season. This should be acknowledged in the analysis and discussion.

Recommendation: Document ambient temperature and water temperature at each visit to enable seasonal analysis and comparison with any future monsoon-period studies.

— End of Protocol V7 —

